POKHARA UNIVERSITY

Level: Bachelor | Semester – Fall | Year : 2012
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Turing test. Machine can be made intelligent artificially but ultimately human make the machines. So who is more intelligent – the artificial machine or the person? Discuss. 4+4

   b) In a crypto-arithmetic puzzle, the variable A, B, C, D, E and F can take values from 1 to 7. The variables must all be different and, when taken as digits, they must satisfy the following sum. Solve the following problem as CSP. 7

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  AB
+ CD
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  EF
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2. a) Explain production systems with the help of 8-puzzle example. 8

   b) What are the problems that may arise in hill climbing searching? How they can be handled? Explain. 4+3

OR

   a) You are given two jugs, a 4 liters gallon one and a 3 liters gallon one. Neither have any measuring markers on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 liters of water into the 4 liters gallon jug? Solve this problem with state space representation. 8

   b) Explain Min-Max search strategy with example. 7

3. a) What is meaning of word ***heuristics*** in the context of search strategies? What conditions on A* search is required to guarantee completeness and optimality. 3+4

   b) What are the different dimensions of search strategy? Compare the 4+4

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BFS and DFS with different dimension.

4 a) The logical operator "$\leftrightarrow$" is read "if and only if". $P \leftrightarrow Q$ is defined as being equivalent to $(P \rightarrow Q) \wedge (Q \rightarrow P)$. Based on this definition, show that $P \leftrightarrow Q$ is logically equivalent to $(P \vee Q) \rightarrow (P \wedge Q)$ using truth table.

b) Represent the following using first order Logic.

i) Prasant likes easy courses.
ii) Science courses are hard.
iii) All the courses in the computer departments are easy.
iv) AI is a computer course.
Also, use **resolution** to answer the question, "What course would Prasant like?"

OR

a) Prove that " jeevan is happy" with the help of following facts expressed in CNF.

i. $\neg$pass(X,history) V $\neg$win(X,lottery) V happy(X)
ii. $\neg$study(X) V pass(Y,Z)
$\neg$lucky(W) V pass(W,U)
iii. $\neg$study(jeevan)
lucky(jeevan)
iv. $\neg$lucky(U) V win(U,lottery)

b) Explain with Example, how first order logic sentences are converted into conjunctive normal form (CNF). 7

5. a) How can you say FOPL extend the expressiveness of PL?

b) Explain the forward and backward representational mapping with diagram. 5

c) Explain Semantic Network with suitable example. 5

6. a) Explain the basic components of expert system. How can we make expert system knowledge base reusable? 5

b) What do you mean by learning by analogy? Explain derivational analogy. 4

7. Write short notes on **any two:** 3×

a) Bayesian Network

b) Frames 2×

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